

KIDNEY AS A FILTRATION SYSTEM

3.2 Kidney as a Filtration System

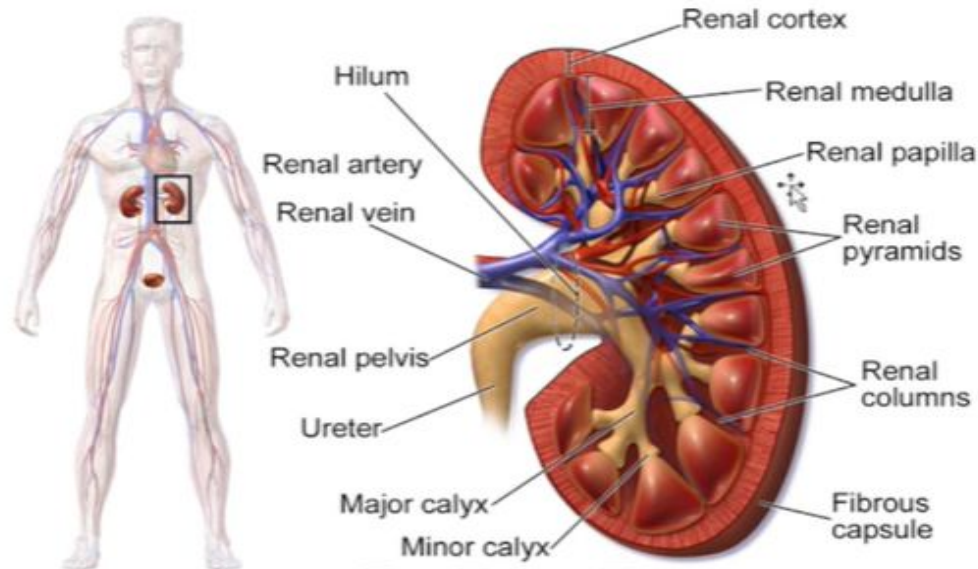


Figure: Anatomy of kidney

The kidney is a complex organ that acts as a filtration system for the body. It removes waste and excess fluid from the bloodstream and maintains a delicate balance of electrolytes, hormones, and other substances that are critical for the body's normal functioning.

The kidney also plays an important role in regulating blood pressure by secreting the hormone renin, which helps control the balance of fluid and electrolytes in the body. It also regulates red blood cell production and the levels of various minerals in the blood, such as calcium and phosphorus.

Without the kidney, waste and excess fluid would accumulate in the body, leading to serious health problems.

3.2.1 Architecture of Kidney

The kidney is composed of functional units called nephrons, which are the basic structural and functional units of the kidney. Each kidney contains approximately one million nephrons, and each nephron performs the functions of filtration, reabsorption, and secretion.

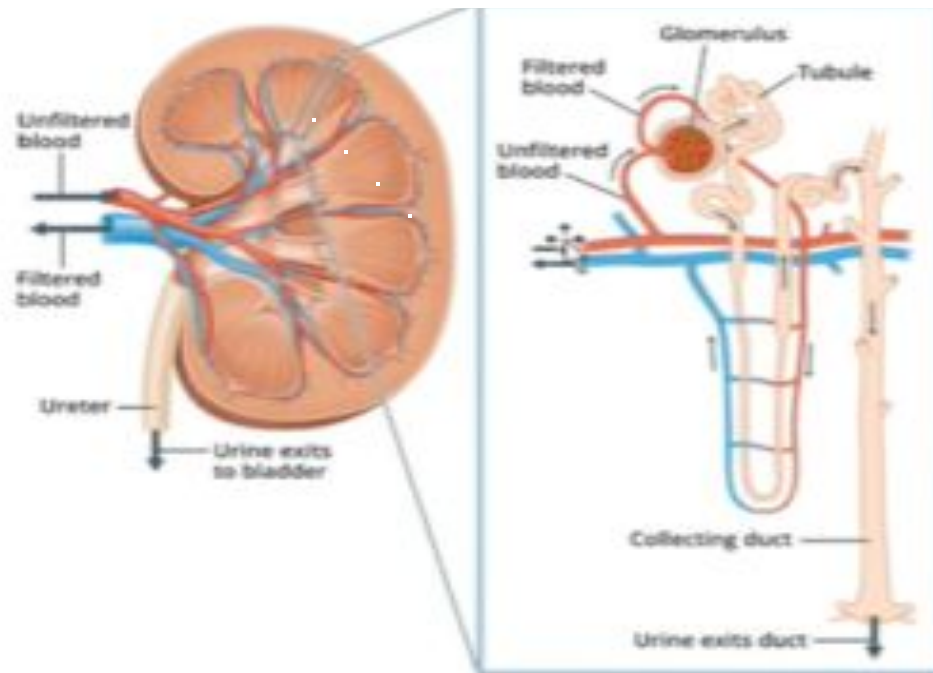


Figure: Representing kidney and nephron

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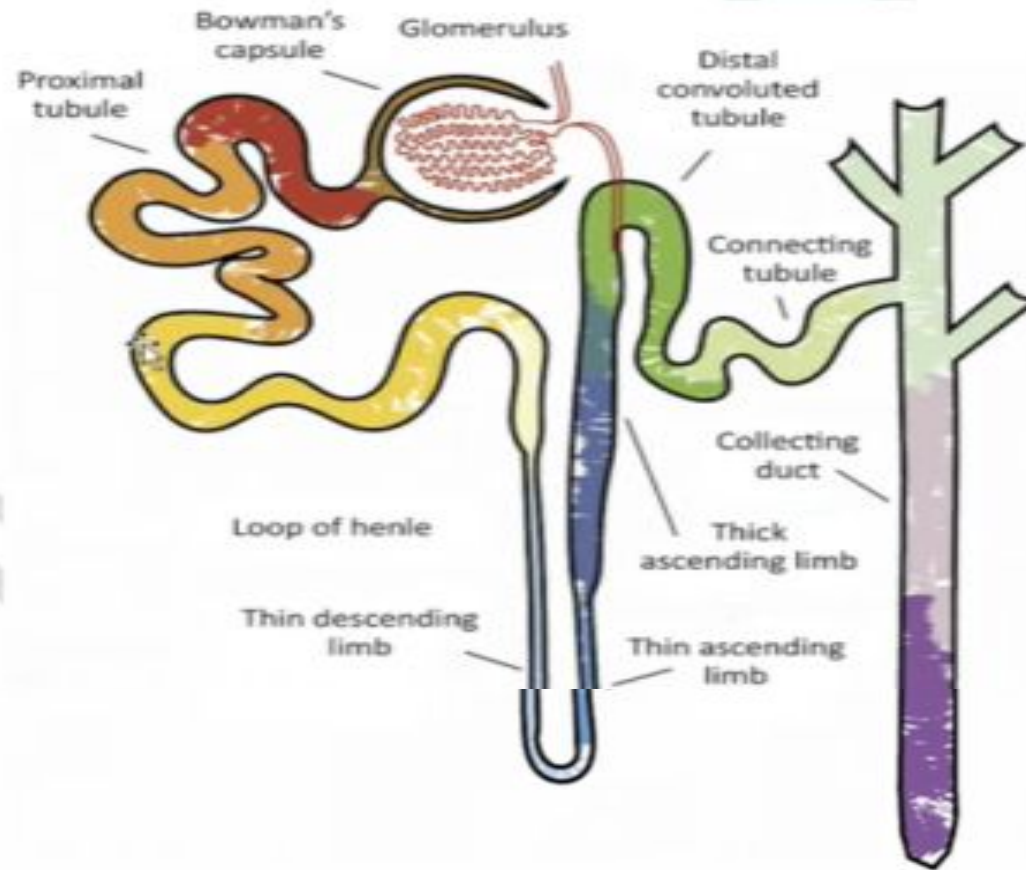


Figure: Representing the parts of nephron

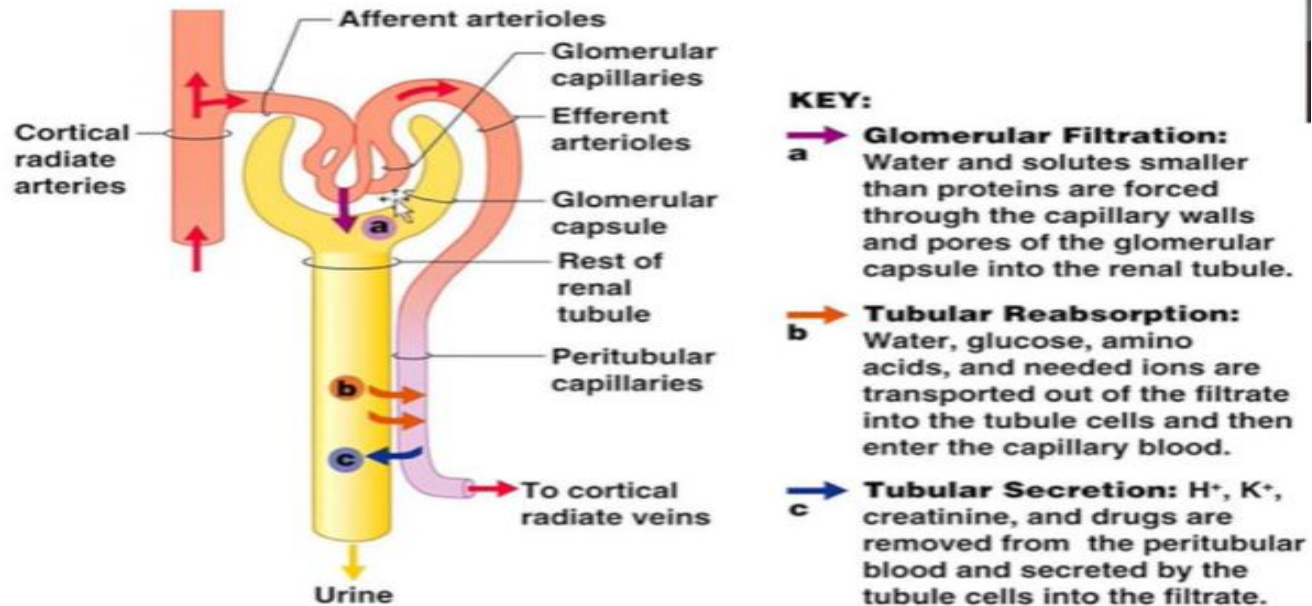
The nephron is comprised of several key structures:

- Bowman's capsule: This is a cup-shaped structure that surrounds the glomerulus and filters waste and excess fluid from the bloodstream into the renal tubule.

- Glomerulus: A network of tiny blood vessels within the Bowman's capsule that filters waste and excess fluid from the bloodstream.
- Proximal convoluted tubule: A segment of the renal tubule that reabsorbs important substances, such as glucose, amino acids, and electrolytes, back into the bloodstream.
- Loop of Henle: A U-shaped segment of the renal tubule that is critical for the reabsorption of ions and water.
- Distal convoluted tubule: A segment of the renal tubule that regulates the levels of electrolytes and other important substances in the bloodstream.
- Collecting duct: A series of ducts that collect the filtrate from the renal tubules and transport it to the renal pelvis, where it drains into the ureter and eventually into the bladder.

The nephrons are surrounded by a network of blood vessels, including the afferent arteriole and the efferent arteriole, which bring blood into and out of the glomerulus, respectively. The filtrate produced by the nephron passes through the renal tubules, where it is modified by reabsorption and secretion, before being eliminated from the body as urine.

MECHANISM OF FILTRATION-URINE FORMATION



The mechanism of filtration in the kidneys is a complex process that involves multiple steps to remove waste and excess fluids from the bloodstream. The following is a summary of the steps involved in the filtration process:

- Blood enters the kidney through the renal arteries and flows into tiny filtering units called glomeruli.
- At the glomerulus, the pressure in the blood vessels causes a portion of the plasma and dissolved substances to filter out and enter a structure called Bowman's capsule.
- In Bowman's capsule, the filtrate is then transferred into the renal tubules, which are the main filtering units of the kidneys.
- In the renal tubules, the filtrate passes through a series of specialized cells, such as proximal tubular cells and distal tubular cells, which reabsorb important substances such as glucose, amino acids, and electrolytes back into the bloodstream.
- At the same time, the renal tubules secrete waste products, such as urea and creatinine, back into the filtrate.
- Finally, the filtered fluid, now known as urine, is transported through the renal pelvis and ureters to the bladder, where it is eventually eliminated from the body.

This process of filtration, reabsorption, and secretion helps to maintain the proper balance of fluids and electrolytes in the body, as well as to remove waste and excess substances.

3.2.3 Chronic Kidney Disease (CKD)

CKD stands for Chronic Kidney Disease. It is a long-term condition in which the kidneys gradually become less able to function properly. It can be caused by a variety of factors, including diabetes, high blood pressure, and other health problems that damage the kidneys.

Symptoms of CKD include fatigue, swelling in the legs and feet, trouble sleeping, and difficulty concentrating. As the disease progresses, it can lead to more serious complications, such as anemia, nerve damage, and an increased risk of heart disease and stroke.

Treatment for CKD may include lifestyle changes, such as eating a healthy diet and exercising regularly, as well as medications to manage symptoms and underlying health conditions. In severe cases, kidney transplant or dialysis may be necessary.

It is important for individuals with risk factors for CKD to get regular check-ups and to talk to their doctor about how to best manage their condition.

3.2.4 Dialysis Systems

Dialysis is a medical treatment that helps to filter waste and excess fluids from the blood when the kidneys are unable to function properly. There are two main types of dialysis systems: hemodialysis and peritoneal dialysis.

Hemodialysis is a procedure that uses a machine to clean the blood. During hemodialysis, blood is removed from the body, passed through a dialysis machine that filters out waste and excess fluids, and then returned to the body. Hemodialysis typically takes place in a hospital or dialysis center, and is typically performed three times a week for three to four hours at a time.

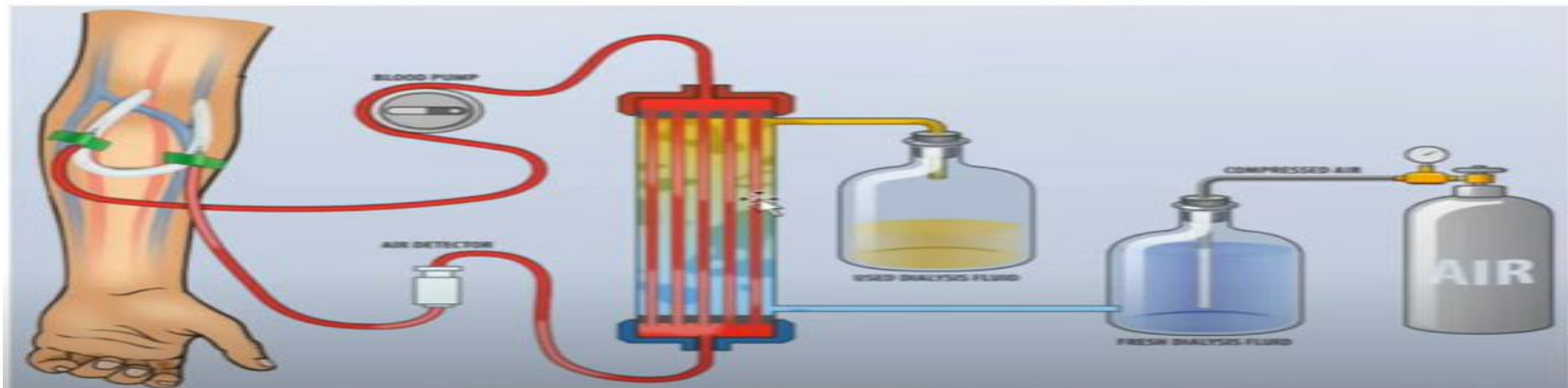
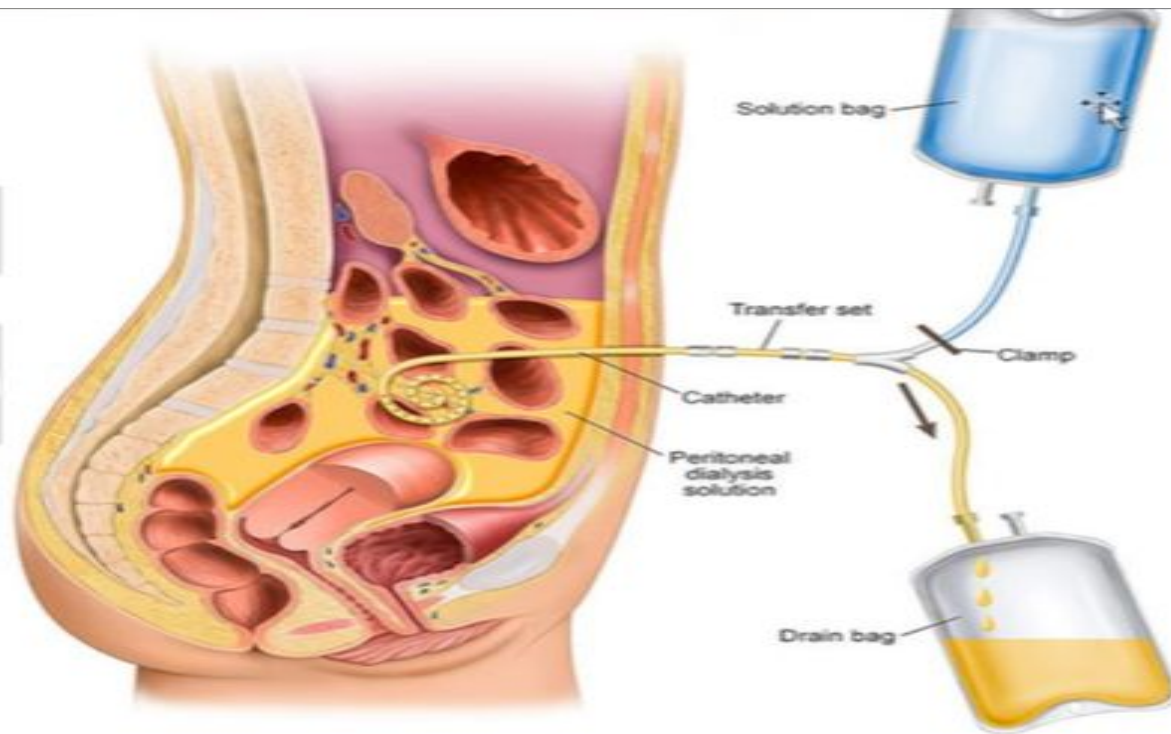


Figure: Representing a Hemodialysis

Peritoneal dialysis is a type of dialysis that uses the lining of the abdomen, called the peritoneum, to filter waste and excess fluids from the blood. A sterile solution is introduced into the abdomen, where it absorbs waste and excess fluids, and is then drained and replaced with fresh solution. Peritoneal dialysis can be performed at home and allows for more flexibility in scheduling.



Both hemodialysis and peritoneal dialysis can effectively treat the symptoms of kidney failure, but each has its own advantages and disadvantages. The choice of dialysis system depends on various factors such as the individual's overall health, lifestyle, and personal preferences.

3.2.5 Artificial Kidney

While much progress has been made in developing an artificial kidney, it is still in the experimental stage and is not yet widely available. Further research and development is needed to improve the efficiency and safety of artificial kidney devices, and to ensure that they can be widely adopted as a treatment for chronic kidney disease.

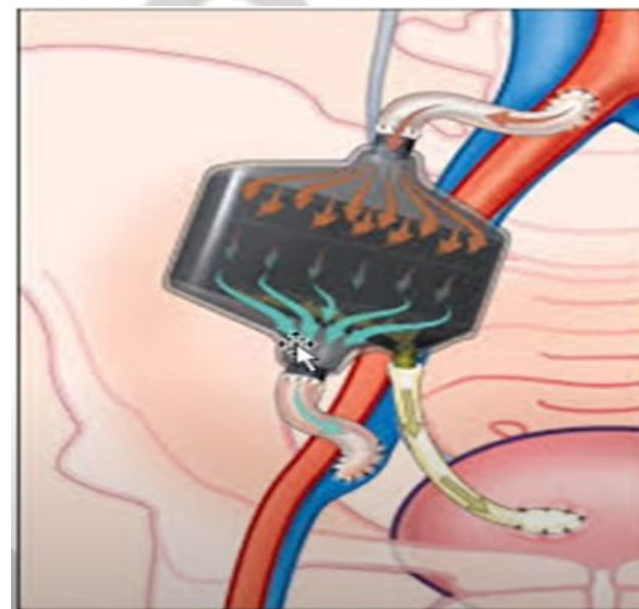


Figure: a) Schematic representation b) a prototype of artificial kidney

An artificial kidney is a device that is being developed to mimic the functions of the human kidney. The goal of an artificial kidney is to provide a more effective and efficient means of treating patients with chronic kidney disease, who currently rely on dialysis or kidney transplantation.

There are currently two main approaches to developing an artificial kidney: a biological approach and a technological approach.

The biological approach involves using living cells, such as kidney cells or stem cells, to create a functional, implantable artificial kidney.

The technological approach involves using synthetic materials, such as silicon or polymer, to create a dialysis device that can filter the blood and remove waste and excess fluids.

It's important to note that while the development of an artificial kidney holds great promise, it is not a cure for chronic kidney disease and patients with kidney failure will still need dialysis or kidney transplantation in the meantime.